



Complete Project Overview and Use Cases

A trip-first workspace for planning, shared accounting, reconciliation, and settlement

Product in one sentence TULIS connects the itinerary to a deterministic, auditable group-expense ledger so a changing trip remains fair, understandable, and settleable.

Project overview

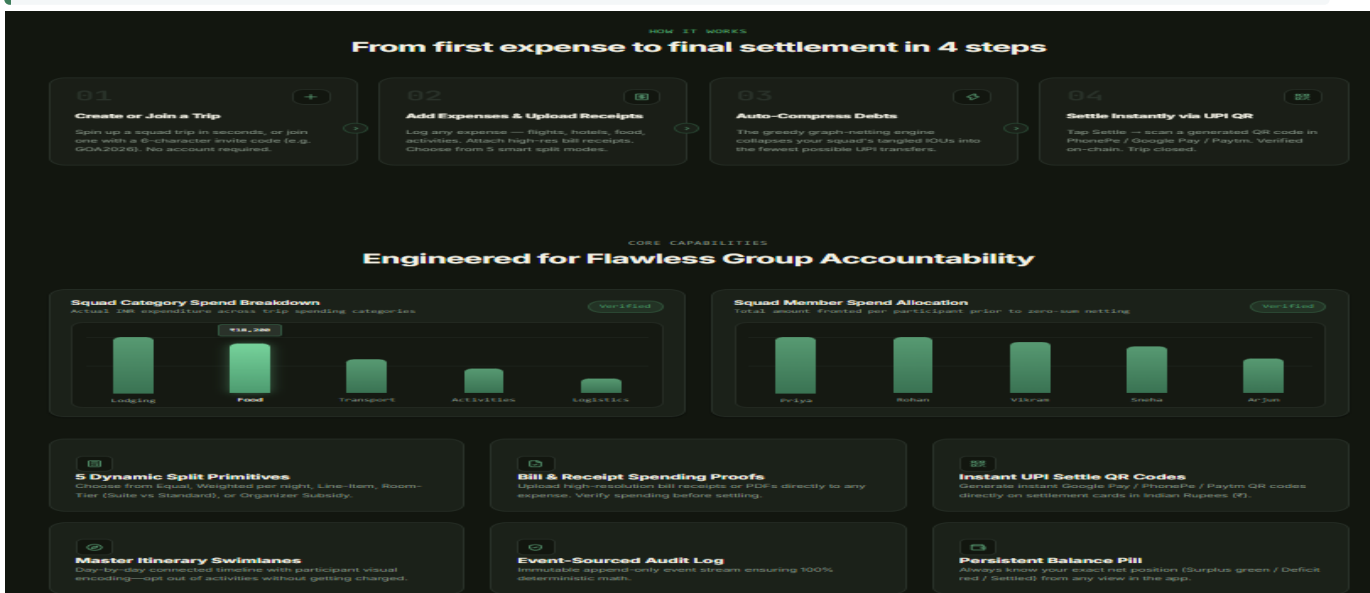
TULIS replaces the fragmented workflow of booking emails, chat messages, spreadsheets, and generic bill splitters with one trip workspace. It combines participants, bookings, activities, expenses, allocations, refunds, payments, anomalies, and settlement history. The financial core is an immutable event-sourced ledger. Balances are derived from events rather than hand-edited, disputed entries can be quarantined, and the system enforces a zero-sum invariant within a ₹0.02 tolerance. This creates a reproducible answer to “who paid, who consumed, who owes, and why?”

Layer	Overview
Workspace	Trip creation, invite links, roles, budgets, currencies, room tiers, UPI details, and private-access controls.
Planning	Bookings, timeline, activity context, room allocation, and schedule/logistics conflict checks.
Accounting	Expenses, six split methods, multi-payer support, cent-safe allocation, debt simplification, disputes, and immutable events.
Change management	What-if scenarios, cancellation policies, proportional refund cascades, duplicate guards, and anomaly detection.
Assistance	Natural-language expense parsing, balance explanations, split recommendations, and deterministic fallbacks.
Settlement	UPI/QR hand-off, minimal payment targets, audit PDF, and RFC 4180 CSV export.
Resilience	Offline command queue, idempotent replay, visible sync status, and reconciliation audit.

Primary use cases

Scenario	How TULIS helps
Friends' multi-day trip	An organizer creates a trip; participants join; different travelers pay for lodging, transport, meals, and activities; TULIS calculates the smallest practical settlement set.
Unequal lodging	Suite, standard, economy, different nights, or partial attendance are handled through room-tier, weighted, line-item, or manual allocation.
Offline travel	Expenses and payments queue locally during poor connectivity and replay safely after reconnection without duplicate events.
Cancellation or drop-out	A full, partial, per-head, or non-refundable policy previews and propagates debt relief with exact cent preservation.
Explainable settlement	Each participant sees an itemized balance explanation, payment target, UPI hand-off, and durable audit record.

Core promise Plan together. Split fairly. Settle with confidence. TULIS is designed for the moments when ordinary splitters become difficult: unequal participation, edits, refunds, disputes, weak connectivity, and the last mile of payment.



Key features and innovations

Capability	What it does	User value
Zero-sum event ledger	Replays immutable events and quarantines disputed or invalid entries.	Trust
Six-way split engine	Equal, weighted, room-tier, line-item, organizer-subsidy, and manual methods with cent-safe rounding.	Fairness
Debt simplifier	Turns complex obligations into a small set of actionable transfers.	Clarity
Refund cascades	Recomputes payer credit and participant relief after cancellations or drop-outs.	Change safety
Anomaly engine	Flags schedule, capacity, budget, roster, allocation, and transit conflicts.	Prevention
AI with guardrails	Parses language and explains balances while deterministic validation controls financial writes.	Speed
Offline-first sync	Queues, replays, deduplicates, and exposes conflicts instead of silently overwriting.	Resilience
UPI settlement	Links outstanding balances to an optional India-localized payment action.	Completion
Audit exports	Generates settlement PDF and structured CSV for post-trip records.	Portability

Competitor comparison

The comparison is based on public product pages, help materials, app-store listings, and company announcements reviewed for the TULIS market analysis. “Not publicly documented” is not proof that a competitor cannot support a capability. TULIS items marked “planned” require implementation and pilot validation.

Feature	TULIS	Common alternatives	How TULIS stands out
Trip-native ledger	Yes	Groups and travel context vary	Bookings, itinerary events, expenses, and refunds share one model
Immutable event history	Yes	Not consistently documented	Every accepted change is attributable and replayable
Refund cascade	Yes	Cancellation tracking varies	Partial refunds and drop-outs recalculate obligations
Six split methods	Yes	Equal/custom/weighted are common	Room tier, subsidy, line item, and manual rules together
Anomaly + logistics checks	Yes	Budget or schedule checks vary	Financial and itinerary conflicts become actionable
Conflict-safe offline replay	Planned	Offline entry is common; guarantees vary	Idempotent queue, sync status, and zero-sum recheck
Explainable AI fallback	Yes	OCR/AI in selected products	AI cannot silently mutate balances
UPI tied to ledger	Planned	Payment support is market-dependent	India-localized path from debt to payment action
Audit PDF + CSV	Yes	Exports vary by product/tier	Post-trip record includes calculations, events, and verification

Differentiation thesis \ Competitors already solve the basic “split a bill” job. TULIS should win the harder “keep a changing trip financially understandable” job by combining travel context, fairness, change management, offline reliability, explainability, and settlement in one auditable workflow.

Launch focus and proof points

Initial segment

Start with India domestic leisure trips among friends and families coordinating multi-day travel. This creates a focused test environment for unequal participation, shared bookings, many small expenses, UPI hand-off, and organizer-led adoption. Enterprise travel-and-expense management is an adjacent future market, not the first positioning.

First-release proof points

1. Organizer creates a trip and invites participants with low friction.
2. A real trip handles unequal rooms, multi-payer expenses, offline entries, a partial refund, and final settlement without manual arithmetic.
3. Every balance can be explained, exported, and traced to source events.
4. Pilots show fewer corrections, clearer refund outcomes, higher settlement completion, and repeat-trip intent than the group’s current workflow.